

Automated Home Folder (H: Drive) Setup for Domain Users

Objective:

To automatically create personal home folders (H: drive) for each domain user under \\DC1\HomeDirs, with NTFS permissions and group membership, using a PowerShell script.

Step-by-Step Guide:

Step 1: Create the Network Share for Home Directories

1. On the server, create a folder (e.g., C:\HomeDirs).
2. Share it as HomeDirs:
 - Right-click > Properties > Sharing > Advanced Sharing...
 - Share name: HomeDirs
 - Permissions: Grant "Domain Users" FULL CONTROL

Setting NTFS Permissions:

- Navigate to the Security tab > Advanced
- Disable inheritance (choose "Convert inherited permissions" when prompted)
- Remove inherited entries like "Everyone"
- Add:
 - Domain Users → This folder only → Read & Execute, List Folder Contents
 - CREATOR OWNER → Subfolders and files only → Full Control

Step 2: Manual Test for One User

1. Open Active Directory Users and Computers (ADUC)
2. Right-click a test user (e.g., jdoe) > Properties
3. Go to the Profile tab
4. Under Home folder, select:
 - Connect: H:
 - To: \\DC1\HomeDirs\%username%
5. Apply > OK
6. Test by logging in as jdoe on a VM and check File Explorer and the original folder C:\HomeDirs

Step 3: Create a Template User in ADUC

1. In your target OU (e.g., Students), right-click > New > User
2. Input details:
 - First name: Template
 - Last name: Student
 - User logon name: template.student
3. Set a basic password (e.g., Student123!) and uncheck "User must change password at next logon"

4. Optional: Uncheck "User cannot change password" and "Password never expires" as needed
5. Configure Home Folder:
 - Profile tab > Home folder:
 - * Connect: H:
 - * To: \\DC1\HomeDirs\%username%
6. Optional: Add to group "Stellar_students" under the Member Of tab

Step 4: Additional Setup for Legacy Users Using PowerShell

Create a script named BulkHomeDriveSetup.ps1:

```
# BulkHomeDriveSetup.ps1
```

```
# Set base directory and student group
```

```
$HomeDirRoot = "\\DC1\HomeDirs"
```

```
$StudentGroup = "stellar_students"
```

```
# List of usernames (sAMAccountName format)
```

```
$UserList = @("crex", "mlan", "jsmith", "adoe", "bchan") # Add more as needed
```

```
foreach ($Username in $UserList) {
```

```
    $User = Get-ADUser -Identity $Username -Properties HomeDirectory
```

```
    if ($User) {
```

```
        $HomePath = "$HomeDirRoot\$Username"
```

```
        # Create folder if not exists
```

```
        if (-not (Test-Path $HomePath)) {
```

```
            New-Item -Path $HomePath -ItemType Directory | Out-Null
```

```
            Write-Output "Created folder for $Username"
```

```
        }
```

```
        # Set NTFS permissions
```

```
        $Acl = Get-Acl $HomePath
```

```
        $AccessRule = New-Object
```

```
System.Security.AccessControl.FileSystemAccessRule("$Username", "FullControl",  
"ContainerInherit, ObjectInherit", "None", "Allow")
```

```
        $Acl.SetAccessRuleProtection($True, $False) # Disable inheritance
```

```
        $Acl.SetAccessRule($AccessRule)
```

```
        Set-Acl -Path $HomePath -AclObject $Acl
```

```
        Write-Output "Set permissions for $Username"
```

```
        # Update AD user object with home directory
```

```
Set-ADUser -Identity $Username -HomeDirectory $HomePath -HomeDrive "H:"  
Write-Output "Assigned H: drive to $Username"
```

```
# Add to student group  
Add-ADGroupMember -Identity $StudentGroup -Members $Username  
Write-Output "Added $Username to $StudentGroup"  
} else {  
    Write-Warning "$Username not found in AD."  
}  
}
```

 Result:

- For each user:
 - A folder is created: \\DC1\HomeDirs\username
 - They receive Full Control NTFS permissions on their folder only
 - Their AD profile is updated to mount it as H:
 - They are added to the stellar_students group

Notes:

- This script automates folder creation and AD property updates.
- New users should be cloned from a template account.
- Scalable for educational and SME environments.
- If a user was missed, re-run the script with the correct sAMAccountName.